

Mathematiktext-Klassifizierung mittels Wikipediadaten

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Motivation

Ich möchte jegliche Technische notizen automatisch klassifizieren lassen.

Publikumsfrage:

Welchem Mathematik-Themenbereich gehören diese drei Sätze jeweils an?

“... die Zahl 6 ist keine Primzahl, aber die Zahl 5 ...”

“... die Projektionsmatrix P zum Quadrat ist sie selbst, $P^2 = P$...”

“... $F[f](\omega) := \int f(t)e^{-t\omega}dt$ ist auch wieder invertierbar ...”

“... nach dem Satz von Vieta kann man ...”

Demo!

Enter text to search and classify (or hit Enter to use default):

I like coordinate transforms in 3d space

Query-embedding dimension: 768, first few values:

[-0.0261, -0.0253, 0.0130, -0.0107, -0.0045, -0.0051, 0.0760, 0.0084, -0.0363, -0.0228 ...]

Closest embedded math wikipedia pages:

1.	dist = 0.912	Coordinate systems	wikipedia.org/wiki/List_of_common_coordinate_transformations
2.	dist = 0.937	Transformation_(function)	wikipedia.org/wiki/Helmert_transformation
3.	dist = 0.981	Transformation_(function)	wikipedia.org/wiki/Translation_of_axes
4.	dist = 0.982	Transformation_(function)	wikipedia.org/wiki/Transformation_matrix
5.	dist = 1.016	Metric tensors	wikipedia.org/wiki/Curvilinear_coordinates
6.	dist = 1.025	Tensors	wikipedia.org/wiki/Cartesian_tensor
7.	dist = 1.033	Transformation_(function)	wikipedia.org/wiki/Rotation_of_axes
8.	dist = 1.064	Coordinate systems	wikipedia.org/wiki/Quadrant_coordinates
9.	dist = 1.065	Projective geometry	wikipedia.org/wiki/Homogeneous_coordinates
10.	dist = 1.084	Triangle geometry	wikipedia.org/wiki/Trilinear_coordinates
11.	dist = 1.087	Linear algebra	wikipedia.org/wiki/Three-dimensional_rotation_operator
12.	dist = 1.091	Clipping_(computer_graphics)	wikipedia.org/wiki/Clip_coordinates
13.	dist = 1.101	3D computer graphics	wikipedia.org/wiki/Quaternions_and_spatial_rotation
14.	dist = 1.106	Geometry stubs	wikipedia.org/wiki/Planar_projection
15.	dist = 1.115	Geometry in computer vision	wikipedia.org/wiki/Homography_(computer_vision)
16.	dist = 1.123	Geometry in computer vision	wikipedia.org/wiki/Collinearity_equation
17.	dist = 1.124	Geometric algebra	wikipedia.org/wiki/Plücker_coordinates
18.	dist = 1.128	Metric tensors	wikipedia.org/wiki/Tensors_in_curvilinear_coordinates
19.	dist = 1.129	Geometry in computer vision	wikipedia.org/wiki/Trifocal_tensor
20.	dist = 1.131	Computer graphics	wikipedia.org/wiki/Glossary_of_computer_graphics

Wikipedia category counts:

	16.7%	(5 out of 30)	Transformation_(function)
	16.7%	(5 out of 30)	Geometry in computer vision
	13.3%	(4 out of 30)	Coordinate systems
	6.7%	(2 out of 30)	Metric tensors
	3.3%	(1 out of 30)	Tensors
	3.3%	(1 out of 30)	Projective geometry

My custom categorization:

	53.3%	geometry
	10.0%	linear algebra
	6.7%	analysis

Enter text to search and classify (or hit Enter to use default):
my friend gave a presentation on the use of Kalman filters in tax time series

Query-embedding dimension: 768, first few values:

[-0.0596, -0.0727, -0.0187, 0.0384, -0.0251, -0.0032, 0.0437, -0.0015, -0.0450, 0.0173 ...]

Closest embedded math wikipedia pages:

1.	dist = 1.058	Nonlinear_filters	wikipedia.org/wiki/Kalman_filter
2.	dist = 1.078	Nonlinear_filters	wikipedia.org/wiki/Extended_Kalman_filter
3.	dist = 1.095	Time_series_models	wikipedia.org/wiki/State-space_representation
4.	dist = 1.108	Control_theory	wikipedia.org/wiki/Schmidt-Kalman_filter
5.	dist = 1.131	Nonlinear_filters	wikipedia.org/wiki/Switching_Kalman_filter
6.	dist = 1.155	Nonlinear_time_series_analysis	wikipedia.org/wiki/Nonlinear_autoregressive_exogenous_model
7.	dist = 1.155	Stochastic_differential_equations	wikipedia.org/wiki/Filtering_problem_(stochastic_processes)
8.	dist = 1.156	Nonlinear_filters	wikipedia.org/wiki/Bellman_filter
9.	dist = 1.170	Time_series_models	wikipedia.org/wiki/Vector_autoregression
10.	dist = 1.170	Signal_processing	wikipedia.org/wiki/Autoregressive_model
11.	dist = 1.173	Signal_processing	wikipedia.org/wiki/Alpha_beta_filter
12.	dist = 1.179	Finnish_statisticians	wikipedia.org/wiki/Pentti_Saikkonen
13.	dist = 1.187	Canadian_mathematicians	wikipedia.org/wiki/Rogemar_Mamon
14.	dist = 1.215	Time_series	wikipedia.org/wiki/Tracking_signal
15.	dist = 1.216	Time_series_econometricians	wikipedia.org/wiki/Helmut_Lütkepohl
16.	dist = 1.222	Econometric_models	wikipedia.org/wiki/Econometric_model
17.	dist = 1.230	Multivariate_time_series	wikipedia.org/wiki/Bayesian_vector_autoregression
18.	dist = 1.234	Time_series	wikipedia.org/wiki/Berlin_procedure
19.	dist = 1.234	Time_series	wikipedia.org/wiki/Hodrick-Prescott_filter
20.	dist = 1.237	Nonlinear_filters	wikipedia.org/wiki/Auxiliary_particle_filter

Wikipedia category counts:

	23.3%	(7 out of 30)	Nonlinear_filters
	16.7%	(5 out of 30)	Time_series
	6.7%	(2 out of 30)	Time_series_models
	6.7%	(2 out of 30)	Signal_processing
	3.3%	(1 out of 30)	Control_theory
	3.3%	(1 out of 30)	Nonlinear_time_series_analysis

My custom categorization:

	43.3%	probability_stats
	33.3%	information_crypto
	20.0%	mathematicians
	3.3%	analysis

Enter text to search and classify (or hit Enter to use default):

In the world of mathematics, matrices are more than just arrays of numbers; they hold the keys to solving complex problems across various fields, from engineering to economics. But what happens when you need to find the inverse of a matrix? This is where things can get tricky. The concept itself might seem daunting at first glance, but with a little guidance and some handy tools, it becomes much more approachable.

To start off, let's clarify what we mean by 'inverse.' Just like how multiplying a number by its reciprocal gives you one (for example, 2 multiplied by 0.5 equals 1), in matrix terms, finding an inverse means identifying another matrix that will yield the identity matrix when multiplied together. The identity matrix acts as a neutral element in multiplication—think of it as '1' for matrices.

Query-embedding dimension: 768, first few values:

[-0.0271, -0.0440, -0.0145, -0.0105, -0.0310, 0.0328, -0.0019, 0.0132, 0.0102, 0.0133 ...]

Closest embedded math wikipedia pages:

1. dist = 0.680	Determinants	wikipedia.org/wiki/Invertible_matrix
2. dist = 0.826	20th-century_Chinese_mathematicians	wikipedia.org/wiki/Guorong_Wang
3. dist = 0.905	Matrices	wikipedia.org/wiki/Integer_matrix
4. dist = 0.908	Matrices	wikipedia.org/wiki/Generalized_inverse
5. dist = 0.910	Exchange algorithms	wikipedia.org/wiki/Gaussian_elimination
6. dist = 0.910	Matrix theory	wikipedia.org/wiki/Partial_inverse_of_a_matrix
7. dist = 0.918	Numerical linear algebra	wikipedia.org/wiki/Moore-Penrose_inverse
8. dist = 0.935	Linear algebra stubs	wikipedia.org/wiki/Frobenius_matrix
9. dist = 0.963	Linear algebra stubs	wikipedia.org/wiki/Drazin_inverse
10. dist = 0.977	Numerical linear algebra	wikipedia.org/wiki/Inverse_iteration
11. dist = 0.981	Linear algebra	wikipedia.org/wiki/Elementary_matrix
12. dist = 0.981	Lemmas in linear algebra	wikipedia.org/wiki/Woodbury_matrix_identity
13. dist = 0.986	Matrix theory	wikipedia.org/wiki/Sherman-Morrison_formula
14. dist = 0.987	Linear algebra stubs	wikipedia.org/wiki/Unimodular_polynomial_matrix
15. dist = 0.988	Matrices	wikipedia.org/wiki/Jacket_matrix
16. dist = 0.994	Computational number theory	wikipedia.org/wiki/Itoh-Tsujii_inversion_algorithm
17. dist = 1.003	Numerical libraries	wikipedia.org/wiki/Efficient_Java_Matrix_Library
18. dist = 1.006	Experimental mathematics	wikipedia.org/wiki/Inverse_Symbolic_Calculator
19. dist = 1.020	Matrix decompositions	wikipedia.org/wiki/Matrix_decomposition
20. dist = 1.021	Matrix multiplication algorithms	wikipedia.org/wiki/Matrix_multiplication_algorithm

Wikipedia category counts:

	13.3%	(4 out of 30)	Matrices
	13.3%	(4 out of 30)	Linear algebra stubs
	10.0%	(3 out of 30)	Numerical linear algebra
	6.7%	(2 out of 30)	Determinants
	6.7%	(2 out of 30)	Exchange algorithms
	6.7%	(2 out of 30)	Matrix theory

My custom categorization:

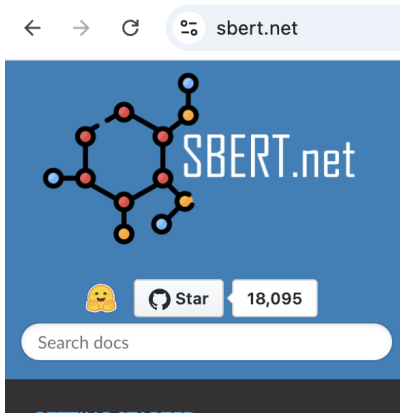
	33.3%	analysis
	30.0%	linear algebra
	13.3%	algos cs
	6.7%	mathematicians
	3.3%	number_theory

Vortragsinhalt

- Suchprogramm-Komponenten
 - Funktionsweise eines **SentenceTransformers**
- **Wikipedia** und meine Datenerstellung
 - Native Wikipedia-Datenstruktur (“Kategorien”, “Mathe-Seite”)
 - Aufbau meines Datenkorpus: heuristiken, filtern, iteration
 - Analyse-Einsichten, “Fun Findings”
- Aussicht, Pläne

Komponenten

- **Text-Einbettungen via Transformer** (BERT by google, SBERT & TU Darmstadt)
- Ähnlichkeitssuche (FAISS by facebook)
 - bestimmt **argmax** für **L2-Vektor-Abstand**, heute nicht tiefer besprochen
- Datensatz (EU's "OpenAIRE"/"wikinformatrics" Projekt und offizielle API)



FAISS

Article [Talk](#)

From Wikipedia, the free encyclopedia

FAISS (**Facebook AI Similarity Search**^[3]) is a library for efficient [search](#) and [clustering](#) of vectors. It contains algorithms for any size, up to ones that possibly do not fit in memory, and [parameter tuning](#).

FAISS is written in [C++](#) with complete wrapper. Some useful algorithms are implemented on the [G](#)

Publikumsfrage:

Für was steht eigentlich...

Chat

G

P

T

Chat

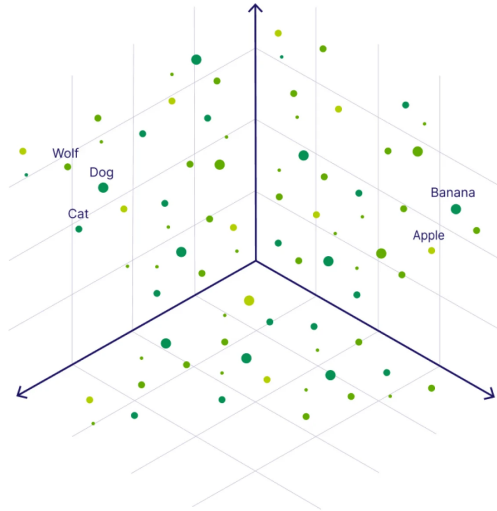
Generative

Pre-trained

Transformer

Startpunkt:

Fixe ursprüngliche Wort-Kodierung in $\mathbb{R}^{786}$, für alle 200k Englischen Wörter.



Startpunkt, links:

Hier repräsentiert jede row von X , links, eine fixe ursprüngliche Wort-Kodierung.

$\mapsto$

Wort	$X_{i,1}$	$X_{i,2}$	$X_{i,3}$	...
Die	-0.41	0.22	-0.67	...
Katze	0.78	-0.33	0.09	...
jagt	0.34	0.47	-0.12	...
hungrig	-0.12	-0.71	0.55	...
die	-0.89	0.15	-0.24	...
Maus	0.56	-0.54	0.31	...

$E(X)_{i,1}$	$E(X)_{i,2}$	$E(X)_{i,3}$	...
-0.47	0.28	-0.62	...
0.72	-0.31	0.19	...
0.10	0.41	-0.09	...
-0.15	-0.78	0.06	...
-0.82	0.14	-0.17	...
0.59	-0.56	0.25	...

“ $\mapsto$ ” repräsentiert die vom eigenen Kontext abhängige Transformer-Abbildung.

Die Einbettung des ganzen Satzes ist (grob) der Mittelwert über die Reihen der Einbettung rechts, also $\text{SatzFeature}_j := \frac{1}{6} \sum_{i=1}^6 E(X)_{ij}$.

$\mapsto$

Wort

Die

Katze

jagt

hungrig

die

Maus

Beispielhaft: Bedeutung $E(X)$

Die

“hungrige, jagende, ... Katze”

“jagt, läuft”

hungrig

die

“müde, ängstliche, ... Maus”

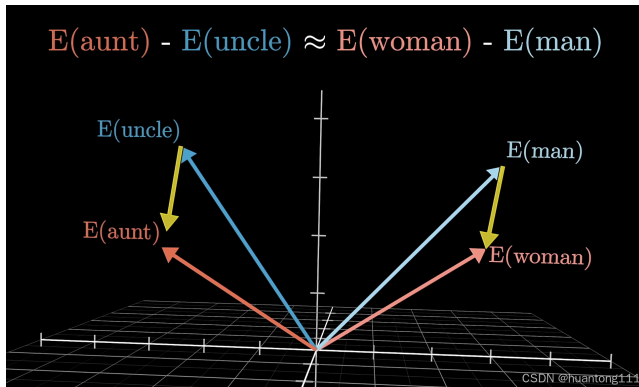
Kontext-Mechanismus:

Jede Zeile von $E(X)$ hängt stark von allen einträgen von X ab.

Training: Im Raum $\mathbb{R}^{786}$ wird mehr kodiert als die 200k Punkte für sich allein!

Rot/Blau: Beispiele für Wort-Einbettungen

Gelb: Richtungen mit **emergent** semantischem Inhalt



Anderes Beispiel. Ein Wort habe ich geändert: **“hungrig” zu “Jerry”**

$\mapsto$

Wort

Die
Katze
jagt
Jerry
die
Maus

Beispielhaft: Bedeutung $E(X)$

Die
“wütende, jagende, ... Katze”
“jagt, läuft”
“Comic-Figur ... Maus ... Jerry”
die
“schlaue, braune, ... Maus”

Kontext-Mechanismus:

Jede Zeile von $E(X)$ hängt stark von allen einträgen von X ab.

	$\mapsto$	
<u>Wort</u>		<u>Beispielhaft: Bedeutung $E(X)$</u>
Apple		"Fruit Apple"
is		is
red		"red color, surface"
<u>Wort</u>	$\mapsto$	<u>Beispielhaft: Bedeutung $E(X)$</u>
Apple		"Apple computer"
USB		"Universal serial bus"
cable		"USB charging cable"

Einbettung von "Apple" wird komplett anderer Vektor, je nach Kontext!

Kennzahlen

- Verwendete Vokabulargröße: **30k**
- Vektorlänge per Wort: **786 dim**
- Max input länge (nativ): **300 Wörter**
- Model weights auf der Platte: **110 Mill. floats / 0.44 GB**

Machbar auf einem Laptop: **3.5k dim; 20k Wörter; 16 GB**

(largest open Facebook model: 16.4k dim; 200k Wörter; 500 GB)

- Gezogen Matheseiten: **40k** (700 MB)
- Einbettung (der ersten 300 Wörter ja): auch **2 Stunden**
- Artikel- und Metadata Pullzeit: **3 Std** (30 GB)
- Datenbank-Erstellung zur Analyse: auch **3 Std**

Wikipedia-Daten

„Warum Wikipedia?‘: offen, groß, verlinkt, Metadaten bereitgestellt.

Englische Wikipedia: ca. **7,1 Mio.** Artikel (gesamt)

Problemstellung: Was ist überhaupt ein “Mathe-Artikel”?

Schlußendlich: Mathe-Artikel-Korpus: 40k Artikel

(i.e., 0.6% der Artikel)

Das Kategorien-Konzept von Wikipedia

Wikipedia hat ein “Kategorien”-Konzept, aber es ist sehr grob:

Auf jede Kategorie kommen circa 3 Artikel! **Über 2 Millionen Kategorien!**

Live erklärung der Kategorien. Startpunkt:

https://en.wikipedia.org/wiki/Benford%27s_law

Für Mathe ist es circa 16, also noch immer 2500 Kategorien.

Allerdings: Man landet schnell außerhalb von Mathe (Geologie, Ökonomie, ...)

pageid=22476294 category=Category:Matrix decompositions
pageid=1087836 category=Category:Matrix normal forms
pageid=16230266 category=Category:Domain decomposition methods
pageid=45081487 category=Category:Matrix multiplication algorithms
pageid=31819845 category=Category:Relaxation (iterative methods)
pageid=30994638 category=Category:Fréchet spaces
pageid=7548377 category=Category:Normed spaces
pageid=7430263 category=Category:Sobolev spaces
pageid=30994686 category=Category:Function spaces
pageid=30994559 category=Category:Metric linear spaces
pageid=10973513 category=Category:Spacetime
pageid=25261497 category=Category:Properties of Lie algebras
pageid=2784704 category=Category:Precession
pageid=34281086 category=Category:Spacecraft attitude control
pageid=71398341 category=Category:Supersymmetric quantum field theory
pageid=3062683 category=Category:Topology of homogeneous spaces
pageid=971666 category=Category:Banach algebras
pageid=32862735 category=Category:Bialgebras
pageid=50604436 category=Category:Composition algebras
pageid=14688386 category=Category:Diagram algebras
pageid=3979846 category=Category:Hopf algebras
pageid=39463033 category=Category:Non-associative algebras
pageid=22974471 category=Category:Operator algebras
pageid=2797309 category=Category:Quaternions
pageid=971791 category=Category:Automorphic forms
pageid=962541 category=Category:Fourier analysis
pageid=18672893 category=Category:Singular integrals
pageid=39316758 category=Category:Coding theorists
pageid=2123467 category=Category:Character encoding

Logik meiner Akquisition

1. Gewählte Mutter-Kategorie: Category:Mathematics
2. Benutzt offene Wikipedia-API mit **Breadth-First Search (BFS)** um Unterkategorien-Kandidaten zu finden
3. Inspeziere manuell
4. Definiere ca **800 keywords** manuell (“algebra”, “group theory”, “mathematician”)
5. Filtere Metadata nach keywords
6. Iteriere die letzten 2 Punkte

Weitere Projektziele

- **Weniger feinmaschige Klassifikation von Texten**
- Erweiterung zu Physik-Artikel
- larger mode: Ließt 6k Wörter
 - Runtime tradeoff?
- Integration in mein Notizbuch-Programm
- Erweiterte Idee: Automatische LLM prompt Generierung mit Mathe-Kategorien-Kontext
 - Design von Quality-Markers für die Artikel? Unterkategorien?

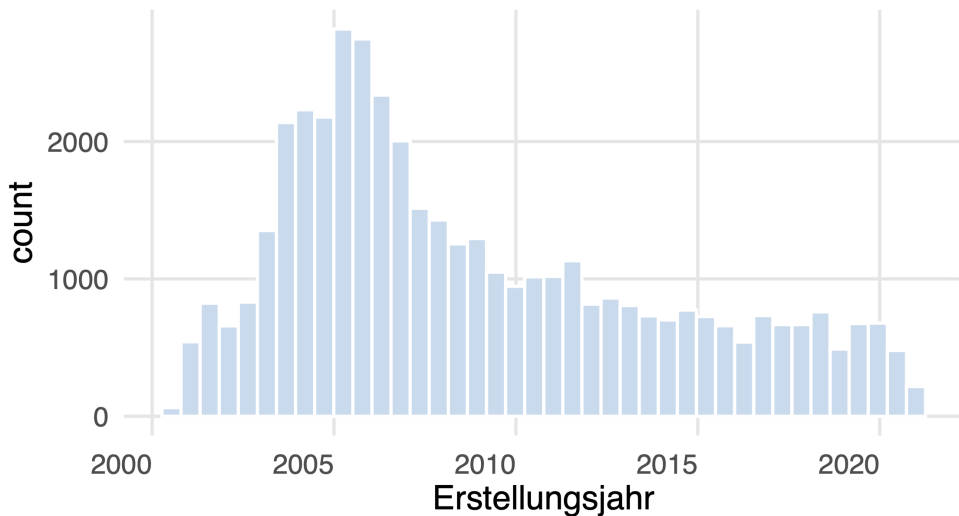
Weitere Projektziele

- Weniger feinmaschige Klassifikation

Für meine weiteren Ziehle habe ich bereits zwei Duzend Überkategorien definiert (“Zahlentheorie”, “Geometrie”, “Statistik”, etc.)

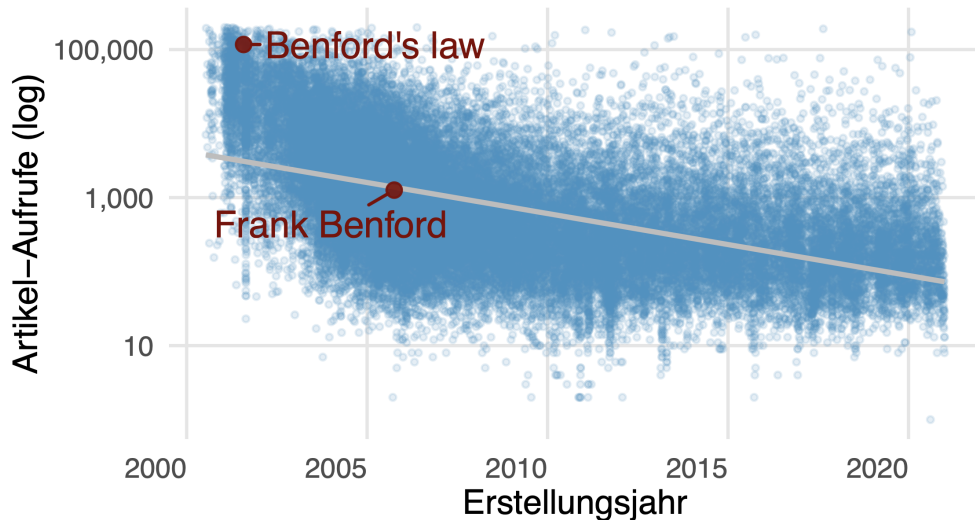
Den Mathe-Datensatz zu verstehen

Histogramm: Artikelerstellungsjahr



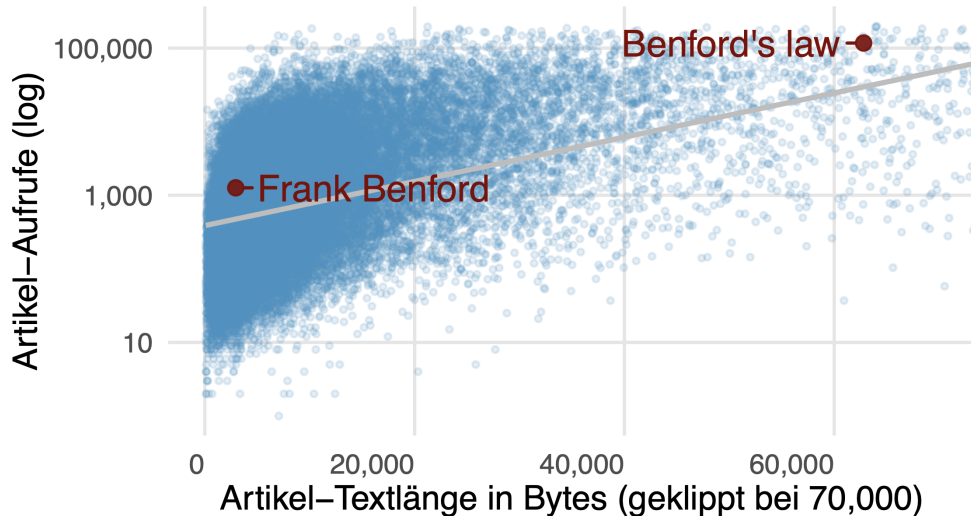
Den Mathe-Datensatz zu verstehen

Artikel–Aufrufe vs Artikelerstellungsjahr



Den Mathe-Datensatz zu verstehen

Artikel–Aufrufe (log) vs Artikel–Textlänge



Den Mathe-Datensatz zu verstehen - Statistik

Table 2: 1,607 pages der von mir Klassifizierten Artikel in probability_stats

n_pages	Titel der Wikipedia Kategorie
128	Stochastic_processes
100	Probability_theorems
99	Regression_analysis
97	Probability_stubs
92	Bayesian_statistics
87	Probability_theory
83	Theory_of_probability_distributions
74	Covariance_and_correlation
71	Monte_Carlo_methods
68	Nonparametric_statistics
NA	...
2	Oceanographic_Time-Series
1	Populated_places_in_the_Municipality_of_Markovci
1	WikiProject_Probability_templates

Den Mathe-Datensatz zu verstehen - Statistik

Table 3: Top 7 and bottom 2 pages in der Überkategorie probability_stats

Aufrufe	page	Hauptkategorie
632588	Normal_distribution	Location-scale_family_probability_distributions
528252	Monty_Hall_problem	Probability_theory_paradoxes
406499	Poisson_distribution	Infinitely_divisible_probability_distributions
289053	Exponential_distribution	Infinitely_divisible_probability_distributions
273238	Pearson_correlation_coefficient	Covariance_and_correlation
241534	Gamma_distribution	Infinitely_divisible_probability_distributions
236237	Logistic_regression	Logistic_regression
NA	...	NA
19	Civic_statistics	Applied_statistics
9	Heyde_theorem	Probability_stubs

Den Mathe-Datensatz zu verstehen - Mathematiker

Table 5: Top 10 and bottom 2 pages in der Überkategorie mathematicians

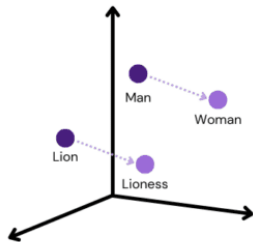
Aufrufe	page	Hauptkategorie
1230395	Ted_Kaczynski	Complex_analysts
1182074	Leonardo_da_Vinci	15th-century_Italian_mathematicians
939498	Alan_Turing	Computability_theorists
670159	Isaac_Newton	History_of_calculus
479642	Socrates	Ancient_Greek_logicians
461580	Aristotle	Ancient_Greek_logicians
444866	Srinivasa_Ramanujan	Indian_combinatorialists
402110	Galileo_Galilei	16th-century_Italian_mathematicians
310478	Confucius	Chinese_logicians
295353	Ada_Lovelace	Ada_(programming_language)
NA	...	NA
3	Simen_Skappel	Norwegian_statisticians
2	Ottaviano_Menni	Italian_mathematician_stubs

Danke!

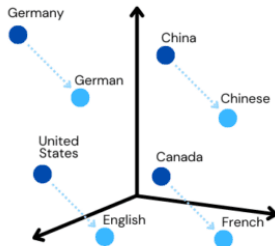
Fragen?

- Zu Anwendungs-Ideen?
- Zu ChatPGT?

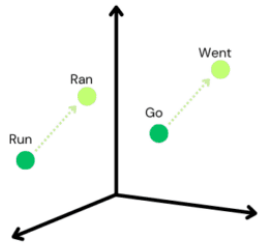
Emergente Bedeutung



Male - Female



Country - Language



Verb Tense

Den Mathe-Datensatz zu verstehen - Statistik

Table 1: Als Statistik klassifizierte Seiten mit 'Law' im Titel (top 5 und bottom 2 bzgl. Artikel-Aufrufe)

Aufrufe	page	Überkategorie	Hauptkategorie
117088	Benford's_law	probability_stats	Theory_of_probability_distributions
103688	Law_of_large_numbers	probability_stats	Asymptotic_theory_(statistics)
72182	Power_law	probability_stats	Theory_of_probability_distributions
33845	Law_of_total_probability	probability_stats	Probability_theorems
32980	Law_of_total_expectation	probability_stats	Algebra_of_random_variables
NA	...	...	...
524	Blumenthal's_zero-one_law	probability_stats	Probability_theory
149	Engelbert-Schmidt_zero-one_law	probability_stats	Probability_theorems